10 April 1959
SUPERSEDING
NAVORD OS 7210
31 August 1956

MILITARY SPECIFICATION

LUG, SUSPENSION, UNDERWATER MINE, MARK 8 MOD 0

1. SCOPE

1.1 This specification establishes the requirements for the procurement of the Lug, Suspension, Underwater Mine, Mark 8 Mod 0, Bureau of Ordnance Drawing No. 1276258. The Lug, Suspension, Underwater Mine, Mark 8 Mod 0 is a non-magnetic eye equipped lug, threaded for coupling with a mine case for suspending the mine from an aircraft release mechanism. The Lug, Suspension, Underwater Mine, Mark 8 Mod 0 is of one type and one class.

2. APPLICABLE DOCUMENTS

2.1 The following standards, drawings, and publications or such portions thereof as designated herein of the issue in effect on the date of invitation for bids form a part of this specification.

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, wood, cleated-plywood
PPP-B-621	Boxes, wood, nailed and locked-corner
PPP-B-636	Boxes, fiber

MILITARY

MIL-A-13374	Adhesive, dextrin
MIL-B-20390	Board, ammunition container

FSC 1350

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWINGS

BUREAU OF ORDNANCE

1276258	Lug, Suspension, Underwater Mine, Mark 8 Mod 0
1476194	Marking Instructions for Storage and Shipping Containers

PUBLICATIONS

BUREAU OF ORDNANCE

OP 400	General Instructions for the Design, Manufacture and Inspection of Naval Ordnance Equipment.
NAVORD OD 10511	Basic Methods for Non-Magnetic Testing With The Test Set Mark 115 Type.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATIONS

Handbook H28	Screw Thread Standards for the Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

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3. REQUIREMENTS

3.1 Description. - The Lug, Suspension, Underwater Mine, Mark 8 Mod 0 is a forged, beryllium copper non-magnetic lug, approximately 3 1/8 inches in diameter and threaded for coupling with a correspondingly threaded recess in a mine case. The Lug, Suspension, Underwater Mine, Mark 8 Mod 0 supports the mine from the release mechanism of a transporting plane by means of an eye in the lug.

3.2 General requirements.

3.2.1 Preproduction samples. - Unless otherwise specified in the contract or order preproduction samples of the Lug, Suspension, Underwater Mine, Mark 8 Mod 0 shall be manufactured using the methods and procedures proposed for the production lot. These samples will be tested as specified in section 4 herein and are for the purpose of determining that the production item meets the requirements of this specification and the design. Provision of samples shall be in accordance with 4.1.2.

3.2.2 Materials. - Unless otherwise specified in the contract, requisition or order all materials entering into the manufacture of the suspension lug shall conform strictly to the detailed requirements indicated on the drawings and specifications listed.

3.2.3 Finishes, coatings, construction and dimensions. - The finish, construction and dimensions for the suspension lug shall be strictly in accordance with the applicable drawings, specifications, and publications listed herein.

3.2.4 Marking. - Marking shall be in accordance with the applicable drawings and MIL-STD-129.

3.2.5 Interchangeability. - The unit, manufactured in strict agreement with the drawing, shall be interchangeable and shall assemble with mating component without selection or fitting. If, during manufacture of the unit, selection or fitting is required and changes in dimensions are necessary, no changes in the part as delineated on applicable drawing will be made without prior revisions and correction of the applicable drawing.

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3.2.6 Gages. - The contractor shall provide himself with whatever gages are necessary and adequate to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawing. If so stipulated in the contract, order, or requisition, the Government will furnish drawings of pertinent Navy Final Inspection Gages for guidance in the design of the contractor's inspection gages. However, such procedure shall not serve to relieve the contractor of his responsibility, in the design and manufacture of such gages as may be required in the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Navy Final Inspection Gages.

3.2.7 Threads. - Unless otherwise specified, the threads of the lug shall be in accordance with the National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings, and the NOT-GO gaging practice shall be as set forth in the aforementioned Handbook H28.

3.2.8 Requests for deviation. - Requests for deviation from this specification, applicable drawings, specifications, publications, materials and processes specified herein, shall be as set forth in OP 400.

3.2.9 Conflicting requirements. - Conflicting requirements arising between this specification or any specifications, publications, or drawings listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation and clarification.

3.3 Definitions. - The following terms will be used throughout this specification on the Lug, Suspension, Underwater Mine, Mark 8 Mod 0 and are here defined for clarity.

3.3.1 Suspension lug. - Reference herein to "lug" or "suspension lug" shall be understood to mean the Lug, Suspension, Underwater Mine, Mark 8 Mod 0.

3.4 Performance requirements and product characteristics. - The Lug, Suspension, Underwater Mine, Mark 8 Mod 0 shall meet the following performance requirements and product characteristics:

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3.4.1 Cleanliness. - At the time of packaging and packing the suspension lug shall be clean and free of corrosion, grease, dirt, metal filings or the like.

3.4.2 Magnetic requirements. - The static magnetic effect of the Lug, Suspension, Underwater Mine, Mark 8 Mod 0 shall not be great enough to cause a total change in flux density of the background magnetic field of more than 0.10 milligauss (peak to trough) when tested in accordance with 4.5.4.1.

3.4.3 Lug surface. - The surface of the suspension lug shall be free of cracks and folds to meet acceptably the test of 4.5.4.2.

3.4.4 Clean-up. - The finished suspension lug shall be thoroughly cleaned of metal chips and other foreign matter. Burrs and sharp edges shall be removed.

3.4.5 Workmanship. - The suspension lug shall be finished in a manner to assure compliance with all requirements of the specification and the drawings with particular attention being directed to neatness in workmanship and marking.

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 Government inspection. - The Government inspector will make such inspections as are necessary to determine that components and assemblies are in accordance with the requirements of the contract, pertinent drawings and specifications. Unless otherwise specified, these inspections will be conducted in accordance with instructions contained in OP 400.

4.1.1 Sampling. - Unless otherwise specified, and when applicable, the sampling plans and procedures used by Government inspector in the determination of the acceptability of products submitted by a supplier for Government inspection shall be in accordance with the provisions set forth in MIL-STD-105.

4.1.2 Preproduction sample. - Unless otherwise specified for each contract or order, a representative sample of ten (10) suspension lugs shall be manufactured using the methods and procedures proposed for the production lot. These shall be submitted to the Government inspector for

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preproduction tests at an activity designated in the contract or by the Bureau of Ordnance, for purposes of determining compliance with the requirements of the contract, specifications, and drawings. Further production of the item by the contractor prior to approval of the contracting activity or the completion of tests on preproduction samples shall be at the contractor's risk. Preproduction samples accepted will be applied as part of the quantity specified by the contract or order.

4.1.3 Lot size. - As applied to Government inspection of units of product the term "lot" shall mean 'Inspection lot", i.e., a collection of units of product submitted by a supplier for Government inspection. Unless otherwise specified, the number of units of product in "inspection lots" will be as determined by the Government inspector and may differ from the quantity designated in the contract or order in a lot for production, shipment or other purpose.

4.2 Contractor's inspection. - Contractor's inspection shall be conducted in accordance with provisions of "Inspection by Contractor" as contained in OP 400.

4.3 Classification of tests. - The inspection and testing of the Lug, Suspension, Underwater Mine, Mark 8 Mod 0 shall be classified as follows:

4.3.1 Acceptance tests. - These tests are to be accomplished on the suspension lugs being submitted for acceptance under contract. Inspection and test procedures shall be submitted for Government approval prior to commencement of production. Acceptance tests shall either be performed by the manufacturer and witnessed by the Government inspector, or shall be conducted by the Government inspector. These tests are defined in 4.5.

4.3.2 Preproduction tests. - Preproduction tests are those which are accomplished on samples selected as specified in 4.1.2 which are representative of the production of the suspension lug after the award of the contract to determine that the lot and production meet the requirements of this specification. These are detailed in 4.5. Acceptance shall be based on no defects in a sample. Failure of a sample to comply with these requirements will result in the rejection of the lot or the cessation of production as determined by the procuring activity.

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4.4 Test equipment. - The following items of test equipment are required to perform the acceptance tests set forth in this specification:

(a) Test Set Mark 115 Mod 4 (Government Furnished)
(b) Pressure chamber suitable for Zyglo tests of 4.5.4.2

4.5 Acceptance tests.

4.5.1 The Contractor or Government inspector shall conduct acceptance tests as specified in 4.3.1 to assure that the Lug, Suspension, Underwater Mine, Mark 8 Mod 0 is in compliance with the requirements of 3.2 and 3.4.

4.5.2 Test conditions. - Unless otherwise specified the Lug, Suspension, Underwater Mine, Mark 8 Mod 0 shall be subjected to acceptance tests under the following conditions:

4.5.2.1 Temperature: Room ambient 65 degrees F to 95 degrees F (18.33 degrees C to 35.0 degrees C).

4.5.2.2 Altitude: Normal ground.

4.5.3 Test and inspection equipment and facilities. - With the exception of the Government furnished equipment specified in 4.4, the manufacturer shall furnish and maintain all necessary test equipment, facilities and personnel for performing all acceptance tests. The test equipment shall be adequate in quantity, and when definite requirements are not specified they shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

4.5.4 Test procedure.

4.5.4.1 Magnetic test. - The magnetic testing of the Lug, Suspension, Underwater Mine, Mark 8 Mod 0 shall be conducted in accordance with Test Method I of OD 10511 with the test face being the flat side of the lug. The distance from the center of the magnetometer to the test face of the lug shall be 2 1/2 ± 1/16 inch. The rate the suspension lug shall be rotated when conducting the test of orientation (C) shall be between 8 and 15 seconds per revolution. When tested as prescribed, a suspension lug producing a magnetic signal exceeding 0.10 milligauss, when measured as described in OD 10511, shall be regarded defective.

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4.5.4.2 Lug surface test. - Prior to testing, each suspension lug shall be thoroughly cleaned to leave all surfaces of the lug completely free of foreign matter including the cleaning agent. The cleaned lug, thoroughly dry, shall then be placed within a suitable pressure container and submerged in Zyglo penetrant, type ZL-1A manufactured by Magnaflux Corporation, Chicago, Illinois, or equivalent. Close the container and apply, at ambient temperature, a 50 pounds per square inch air pressure to the surface of the penetrant for at least 30 minutes. After reducing the pressure and removing the lug, clean the surface of the lug thoroughly of all penetrant and dry. Carefully examine all surfaces of the lug under an untraviolet light for the presence of cracks or folds in the metal. The presence of such cracks or folds shall be cause for regarding the lug defective.

4.5.5 Acceptance criteria. - Failure of the suspension lug to meet any of the requirements and tests of this specification shall be considered cause for regarding the lug defective.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A.

5.1.1.1 Cleaning and drying. - (Not applicable)

5.1.1.2 Preservative application. - (Not applicable)

5.1.1.3 Unit packaging. - Each Suspension Lug Mark 8 Mod 0 shall be individually packaged in a Style OPF fiberboard box conforming to the requirements of Federal Specification PPP-B-636, Class 2, W5c or W5s compliance symbol.

5.1.1.3.1 Prior to unit packaging, a paper tube support, fabricated from ammunition container board material conforming to the requirements of Military Specification MIL-B-20390, consisting of one or more multi-ply or spirally or convolutely wound tubes, bonded together with Dextrin, Military Specification MIL-A-13374, shall be utilized to provide physical protection to threaded end of suspension lug. Paper tube support shall have a 1/4 minimum wall thickness and dimensioned to fit snugly over threaded end of lug. Fiberboard filler pads shall be provided within unit carton as required for a tight pack.

5.1.2 Level B. - (Not applicable)

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5.1.3 Level C.

5.1.3.1 Suspension Lugs Mark 8 Mod 0 shall be packaged in a manner which will afford adequate protection from deterioration and physical damage during shipment from the contractor or supply source to the first receiving activity for immediate use.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior containers. - Unless otherwise specified by the procuring agency, suspension lugs, packaged as specified under paragraphs 5.1.1.3 and 5.1.1.3.1, shall be packed in multiples of two (2), not to exceed a gross shipping weight of seventy (70) pounds.

5.2.1.2 An overseas type wood-cleated plywood box conforming to the requirements of Federal Specification PPP-B-601 or in a Class 2, Style 2, 2 1/2, 3 or a nailed wood box conforming to the requirements of Federal Specification PPP-B-621, shall be employed for the shipping container.

5.2.2 Level B. - (Not applicable)

5.2.3 Level C.

5.2.3.1 Suspension Lugs Mark 8 Mod 0 shall be packed in a manner which will provide protection from damage during direct shipment from the contractor or supply source to the first receiving activity for immediate use. The type of containers used shall conform to applicable carrier rules and regulations.

5.3 Marking.

5.3.1 Normal marking. - In addition to any special marking required by the contract or order, all marking shall be in accordance with Military Standard MIL-STD-129.

5.3.1.1 For details on marking requirements, see Bureau of Ordnance Drawing No. 1476194.

6. NOTES

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6.1 Ordering data. - Procurement documents should specify the desired level of packaging and packing. Level C should not be selected unless it is definitely known that the equipment will be used immediately upon receipt.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any invention that may in any way be related thereto.

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